

Proposed Solution

Date	17 September 2026
Team ID	SWTID-2026-2311
Project Name	AI StudyBuddy API
Maximum Marks	2 Marks

Proposed Solution

Element	Description
Project Title	AI StudyBuddy – AI-Powered Learning Assistance & Study Support Backend
Objective	Build a secure, RESTful Node.js/Express backend that lets an authenticated student upload study material and receive AI-generated (Google Gemini) summaries, flashcards, quizzes, and personalized study plans derived from that material.
Scope	Student and Administrator authentication (JWT access/refresh + bcryptjs + RBAC), study-material upload and storage, and four Gemini-backed AI-generation endpoints (summary, flashcards, quiz, study plan). The React.js client consumes these APIs; the API itself is validated through Postman/Thunder Client.
Problem it Solves	Removes the manual burden of organizing scattered study material, removes the repetitive labor of hand-building flashcards/quizzes, and adds a personalized AI planning layer that no manual method offers.
Proposed Solution / Approach	A modular Express application (controllers / middleware / models / routes / utils) backed by MongoDB Atlas via Mongoose, JWT-protected routes with RBAC, and a dedicated utils/gemini.js that isolates all Gemini AI calls from the rest of the business logic.
Resource Requirements	Node.js v16+, npm v8+, Express.js, MongoDB Atlas cluster, Mongoose, jsonwebtoken , bcryptjs , multer , Google Gemini API key, VS Code + Postman/Thunder Client, Git & GitHub, 4-member team over the 15-29 September 2026 cycle.